

Edge-Preserving Macro-Financial Signal Extraction for Real-Time U.S. Sector Rotation

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July 7, 2026

Abstract

This paper studies whether edge-preserving filters improve real-time extraction of macro-financial regime information in an implementable U.S. sector-allocation strategy. The state vector comprises four public market-based indicators: the VIX, the Russell 2000-to-S&P 500 ratio, the consumer discretionary-to-staples ratio, and the financials-to-utilities ratio. The investable universe consists of the nine Select Sector SPDR ETFs. Six real-time signal-extraction methods are compared: the raw panel, Savitzky-Golay, Perona-Malik anisotropic diffusion, a recursively estimated Hodrick-Prescott trend, endpoint LOESS, and a local-level Kalman filter. The filtered indicators are standardized with trailing 48-month z -scores, aggregated into a composite macro score, classified into expansion, slowdown, and stress regimes using expanding quantile thresholds, and mapped into a monthly long-only allocation to the sector with the highest trailing 60-month regime-conditional mean return. Over the common out-of-sample period from February 2004 through July 2026, the raw strategy delivers a 5.60% annualized return with a Sharpe ratio of 0.304. Savitzky-Golay raises those figures to 13.78% and 0.707, while Perona-Malik delivers 15.85% and 0.820, outperforming the other real-time smoothers. Relative to the raw timing rule, Perona-Malik yields a 9.50 percentage-point annualized mean return spread with a paired t -statistic of 3.38 and positive moving-block and stationary-bootstrap confidence intervals. Relative to the equal-weight sector benchmark, the gain is economically positive but less decisive under conventional two-sided tests. The results therefore support a strong claim that edge-preserving filtering improves the information content of the raw regime signal and a more qualified claim relative to passive sector exposure. Multiple-testing diagnostics based on White's Reality Check, Hansen's Superior Predictive Ability test, the Deflated Sharpe Ratio, and the Probability of Backtest Overfitting reinforce that interpretation.

Keywords: real-time filtering, regime-aware allocation, Savitzky-Golay filter, Perona-Malik diffusion, sector rotation, ETFs, data snooping

1 Introduction

Monthly regime-allocation models face a familiar empirical tension. Macroeconomic and macro-financial indicators are noisy at the horizon relevant for tactical allocation, yet economically meaningful state changes are often abrupt. Excessive smoothing can suppress transitory variation, but it can also delay or attenuate the turning points that matter for portfolio choice. In such settings, the signal-extraction method is part of the economic specification of the strategy rather than a neutral preprocessing detail.

The analysis is conducted in a fully implementable U.S. sector-rotation setting. The state vector is built from four public market-based indicators: the VIX, the Russell 2000-to-S&P 500 ratio, the consumer discretionary-to-staples ratio, and the financials-to-utilities ratio. The investment universe consists of the nine sector SPDR ETFs. The empirical question is whether edge-preserving filters improve the real-time extraction of regime information from that state vector relative to both the unsmoothed signal and more standard benchmark smoothers.

The core comparison is among six signal-extraction methods designed for real-time use: the raw panel, Savitzky-Golay (Savitzky and Golay, 1964), Perona-Malik anisotropic diffusion (Perona and Malik, 1990), a recursively estimated Hodrick-Prescott filter (Hodrick and Prescott, 1997), endpoint LOESS (Cleveland, 1979), and a local-level Kalman filter (Kalman, 1960). All filters are implemented one-sided, the composite macro score is standardized with trailing information only, and sector weights are formed from regime-conditional return estimates available at the formation date. The objective is therefore not to introduce a new predictor, but to evaluate whether better real-time state extraction improves the portfolio consequences of a fixed regime-allocation design.

The results are economically material. Over the common out-of-sample window from February 2004 through July 2026, the raw regime strategy earns a 5.60% annualized return with a 0.304 Sharpe ratio. Savitzky-Golay raises those figures to 13.78% and 0.707. Among the real-time filters considered, Perona-Malik delivers the strongest overall empirical performance, with a 15.85% annualized return and a 0.820 Sharpe ratio. LOESS reaches 12.47% and 0.638, Kalman 8.85% and 0.485, and recursive HP 9.72% and 0.502. The equal-weight sector benchmark remains an important reference point at 10.57% and 0.739. The evidence therefore supports a strong claim relative to the raw timing rule and the alternative benchmark smoothers, but only a more qualified claim relative to passive sector exposure.

The statistical evidence follows the same ranking. Perona-Malik minus raw delivers a 9.50 percentage-point annualized mean spread, a paired t -statistic of 3.38, and positive moving-block and stationary-bootstrap confidence intervals. Against the equal-weight benchmark, the annualized mean spread is 5.56 percentage points. That difference is economically meaningful and its one-sided dependence-robust bootstrap p -values remain small, but the conventional two-sided paired t -test is less decisive. The empirical results therefore support a clearer claim about *signal improvement* than about *benchmark dominance*.

To address data snooping directly, the analysis also evaluates a prespecified family of 249 candidate rules spanning panel definitions, regime-standardization windows, training windows, portfolio

breadth choices, and local parameter grids for Savitzky-Golay and Perona-Malik. Relative to the raw baseline, the family-level White Reality Check and Superior Predictive Ability tests yield p -values of 0.062 and 0.076, respectively. These values are best interpreted as marginal rather than definitive, but they still provide suggestive evidence that the observed improvement is unlikely to be solely attributable to unrestricted model search. Relative to the equal-weight benchmark, the corresponding p -values are substantially weaker. Deflated Sharpe ratio and probability-of-backtest-overfitting diagnostics point in the same direction: the edge over the raw signal survives a meaningful multiple-testing penalty more convincingly than the edge over passive sector exposure.

The mechanism evidence is also informative. Perona-Malik approximately halves the frequency of regime switches relative to the raw signal, reduces high-frequency spectral power materially, and does so without the extreme inertia induced by the recursive HP benchmark. Economically, fewer spurious regime changes should improve performance because they reduce unnecessary turnover and make regime-conditional return estimates more stable. The method therefore appears to improve the state estimate by dampening short-horizon regime noise while preserving medium-horizon cyclical structure well enough to remain actionable in allocation.

The contribution is twofold. First, the paper provides a real-time comparison of edge-preserving and standard smoothers in a regime-aware allocation problem using investable sector ETFs. Second, it shows that the filtering layer can change economic conclusions even when the predictor set and portfolio rule are otherwise held fixed. The contribution is therefore methodological as much as economic, which is why the analysis emphasizes one-sided implementation, benchmark discipline, and explicit multiple-testing controls.

The scope of the evidence should nevertheless be kept in view. The empirical design is confined to U.S. sector ETFs, the evaluation sample contains 270 monthly observations, and the top-one allocation rule is used primarily as a device for isolating the information content of the signal rather than as a claim of portfolio optimality. Those restrictions do not negate the main result, but they do limit the breadth with which it should be generalized.

The rest of the paper proceeds as follows. Section 2 places the study in the macro-finance and signal-extraction literatures. Section 3 describes the data. Section 4 details the real-time filters, the parameter-selection rule, the regime-classification design, and the inference procedures. Section 5 reports the main results, the model-confidence diagnostics, and the robustness evidence. The appendix reports the broader design-search summary, bootstrap-sensitivity checks, turnover-difference tests, and parameter-sensitivity tables.

2 Related Literature

The analysis is most closely related to research linking business conditions to expected returns and cross-sectional leadership. Fama and French (1989) show that expected stock and bond returns vary systematically with changing economic conditions. In a sector-allocation setting, that insight motivates the use of regime-sensitive signals rather than unconditional average-return rankings.

The study is also related to the broader predictive-return literature, where out-of-sample instability and benchmark sensitivity are central concerns. [Welch and Goyal \(2008\)](#) document the fragility of many equity-premium predictors, while [Campbell and Thompson \(2008\)](#) show that economically disciplined forecasting restrictions can improve out-of-sample performance. In parallel, [Ferson and Harvey \(1991\)](#) emphasize that expected returns and risk premia may vary systematically with economic conditions. The present paper does not seek to add another predictive variable to that literature; instead, it examines whether the extraction of the state variable itself can be improved in real time.

The paper further relates to the benchmark discipline emphasized by [DeMiguel et al. \(2009\)](#). Even parsimonious allocation rules must be evaluated against transparent passive alternatives once estimation error, trading activity, and implementation constraints are taken seriously. For that reason, the empirical analysis reports both differences relative to the raw timing rule and differences relative to an equal-weight sector benchmark defined on the same ETF opportunity set.

Finally, the analysis is adjacent to trend-following and time-series signal extraction in asset allocation. For example, [Moskowitz et al. \(2012\)](#) show that relatively simple return-based trend signals may contain economically meaningful information across assets. The present study differs in two respects. First, the signal is constructed from a macro-financial state vector rather than from past asset returns alone. Second, the core object of interest is not momentum per se, but the consequences of alternative real-time filtering methods for regime classification and subsequent portfolio choice.

The distinguishing element of the present analysis lies, therefore, in the signal-processing layer. Savitzky-Golay and Perona-Malik are established methods in engineering and image analysis, but they are not standard tools in empirical asset allocation. Their relevance here follows from the structure of the problem: the object to be estimated is a regime variable that should react to economically meaningful discontinuities while remaining robust to short-run noise. For that reason, the empirical design compares those edge-preserving filters not only with the raw signal and a real-time HP trend, but also with two more familiar smoothing benchmarks from statistics and control theory, namely endpoint LOESS and a local-level Kalman filter. The question is therefore whether the gains attributed to Perona-Malik reflect edge preservation specifically, rather than smoothing in general. That question falls naturally within the scope of quantitative finance because it concerns the interface between time-series filtering, real-time inference, and portfolio choice.

3 Data

3.1 Macro state vector

The macro panel contains four monthly series. The VIX measures market-implied stress and is therefore negatively oriented in the composite macro score. The Russell 2000-to-S&P 500 ratio captures small-cap leadership relative to large caps, a pattern typically associated with broader risk tolerance and improving growth expectations. The consumer discretionary-to-staples ratio proxies

for cyclical relative to defensive consumption demand, while the financials-to-utilities ratio captures the relative pricing of pro-cyclical and defensive balance-sheet risk. Taken together, these series form a parsimonious defensive-rotation panel tailored to sector allocation.

All four indicators are available from January 1998 onward except the two sector ratios, which begin with the launch of the SPDR funds in late 1998. Table 1 reports their coverage and summary statistics. The use of market-based macro-financial indicators has two advantages in the present context: real-time availability and a close correspondence between the information set used for regime extraction and the investable assets used in the allocation exercise.

3.2 ETF universe

The return universe is the nine sector SPDR ETFs: materials (XLB), energy (XLE), financials (XLF), industrials (XLI), technology (XLK), consumer staples (XLP), utilities (XLU), health care (XLV), and consumer discretionary (XLY). Monthly simple returns are computed from adjusted close prices from Yahoo Finance. Coverage begins in January 1999 for all nine funds, which gives a substantially longer implementation window than most modern factor ETFs.

The main portfolio sample starts in February 2004. This start date is determined by the data requirements of the strategy: a trailing 48-month window for standardizing the macro score and a trailing 60-month window for estimating regime-conditional mean returns. The resulting common out-of-sample period contains 270 monthly observations through July 2026.

Table 1: Macro-Financial Panel Descriptives

Series	First observation	Last observation	Observations	Mean	Std. dev.	Min	Max
VIX	1998-01-31	2026-07-31	343	20.211	7.754	9.510	59.890
Russell 2000 / S&P 500	1998-01-31	2026-07-31	343	0.501	0.088	0.309	0.639
Consumer discretionary / staples	1998-12-31	2026-07-31	332	0.967	0.225	0.550	1.596
Financials / utilities	1998-12-31	2026-07-31	332	1.443	0.596	0.655	2.859

Table 2: Sector ETF Descriptives

Series	First observation	Last observation	Observations	Mean	Std. dev.	Min	Max
Materials	1999-01-31	2026-07-31	331	0.008	0.060	-0.224	0.247
Energy	1999-01-31	2026-07-31	331	0.009	0.072	-0.344	0.308
Financials	1999-01-31	2026-07-31	331	0.007	0.060	-0.262	0.218
Industrials	1999-01-31	2026-07-31	331	0.009	0.054	-0.186	0.181
Technology	1999-01-31	2026-07-31	331	0.010	0.067	-0.249	0.248
Consumer staples	1999-01-31	2026-07-31	331	0.006	0.036	-0.126	0.105
Utilities	1999-01-31	2026-07-31	331	0.007	0.044	-0.147	0.132
Health care	1999-01-31	2026-07-31	331	0.008	0.041	-0.145	0.126
Consumer discretionary	1999-01-31	2026-07-31	331	0.009	0.055	-0.176	0.189

4 Methodology

4.1 Why edge-preserving filters matter here

The four indicators in the state vector are noisy market-based proxies for cyclical leadership and risk appetite. They contain short-horizon fluctuations that can generate spurious regime switches, but they also exhibit abrupt transitions around major dislocations such as the global financial crisis, the COVID crash, and the inflation shock of 2022. A useful filter for regime-aware allocation must therefore remove high-frequency noise without mechanically flattening the local discontinuities that contain regime information.

Savitzky-Golay and Perona-Malik address that trade-off through distinct mechanisms. Savitzky-Golay preserves local slope and curvature by fitting low-order polynomials over a trailing neighborhood rather than by repeated averaging. That property is attractive in a regime-allocation setting because local slope changes often coincide with economically relevant turning points in cyclical leadership. Perona-Malik is a nonlinear diffusion scheme that smooths strongly in flat neighborhoods and weakly across large gradients. The filter was originally developed for image segmentation, where edges carry most of the structural information; in a macro-financial setting, abrupt changes around crises and regime shifts play an analogous role. The empirical issue is therefore whether those edge-preserving properties improve regime extraction in a strictly real-time allocation design relative to standard benchmark smoothers such as recursive HP, LOESS, and a local-level Kalman filter.

4.2 Economic orientation of the indicators

Before smoothing, the macro series are oriented so that larger values correspond to a stronger cyclical or risk-taking environment. The three relative-strength ratios already have that interpretation. VIX does not, so it is multiplied by -1 . If $x_{j,t}$ denotes indicator j at date t , the oriented panel is

$$\tilde{x}_{j,t} = \begin{cases} -x_{j,t}, & \text{if } j = \text{VIX}, \\ x_{j,t}, & \text{otherwise.} \end{cases} \quad (1)$$

4.3 One-sided Savitzky-Golay smoothing

For the Savitzky-Golay filter, let x_t denote one oriented series and let h denote the trailing window length. At each month t , a polynomial of order p is fit on the historical window $\{x_{t-h+1}, \dots, x_t\}$ by solving

$$\min_{\beta_0, \dots, \beta_p} \sum_{i=0}^{h-1} \left(x_{t-i} - \sum_{k=0}^p \beta_k i^k \right)^2, \quad (2)$$

and the fitted endpoint is retained,

$$\hat{x}_t^{SG} = \hat{\beta}_0. \quad (3)$$

The empirical implementation is one-sided: the polynomial is re-estimated recursively each month and only the last fitted value is used. This avoids the look-ahead bias embedded in symmetric Savitzky-Golay smoothers. Relative to moving-average-style smoothers, the practical appeal is that local turning points can be retained without forcing the signal to react to every transitory fluctuation.

4.4 One-sided Perona-Malik diffusion

Perona-Malik begins from the anisotropic diffusion equation (Perona and Malik, 1990)

$$\frac{\partial u}{\partial \tau} = \nabla \cdot (c(|\nabla u|) \nabla u), \quad c(g) = \exp \left[- \left(\frac{g}{\kappa} \right)^2 \right]. \quad (4)$$

In one-dimensional discrete form, the update becomes

$$u_i^{(n+1)} = u_i^{(n)} + \Delta t \left[c_{i-1/2}^{(n)} \left(u_{i-1}^{(n)} - u_i^{(n)} \right) + c_{i+1/2}^{(n)} \left(u_{i+1}^{(n)} - u_i^{(n)} \right) \right]. \quad (5)$$

Large local gradients therefore receive little diffusion, whereas small local differences are smoothed more aggressively. The filter is re-run each month on the history available up to that date, and only the final filtered endpoint is retained. In economic terms, the construction is designed to suppress nuisance variation within a regime while preserving the sharper state changes that matter for portfolio rotation.

4.5 Additional real-time benchmark filters

Savitzky-Golay and Perona-Malik are the proposed edge-preserving filters. The remaining smoothers are included as benchmark methods because they represent familiar real-time approaches to trend extraction, local nonparametric smoothing, and linear state-space filtering.

Recursive HP filter. The Hodrick-Prescott benchmark is implemented recursively: at each month t , the standard penalized least-squares trend problem is solved using only observations available up to t , and only the endpoint trend estimate is retained. The monthly smoothing penalty is fixed at $\lambda = 129,600$ (Hodrick and Prescott, 1997), which is the conventional high-frequency calibration used to extract low-frequency business-cycle variation in monthly macro-financial applications. HP is therefore a natural benchmark for a familiar linear trend-extraction design.

Endpoint LOESS. At each month t , the LOESS benchmark fits a locally weighted linear regression on the trailing $h = 24$ observations,

$$(\hat{\beta}_{0,t}, \hat{\beta}_{1,t}) = \arg \min_{\beta_0, \beta_1} \sum_{i=0}^{h-1} w_{i,t} (x_{t-i} - \beta_0 - \beta_1 i)^2, \quad (6)$$

where $w_{i,t}$ denotes the standard tricube kernel weight (Cleveland, 1979). Only the endpoint estimate $\hat{x}_t^L = \hat{\beta}_{0,t}$ is retained, ensuring a fully one-sided implementation. LOESS is therefore a useful benchmark for local nonparametric smoothing without imposing a global state-space structure.

Recursive Kalman filter. The Kalman comparator uses a local-level state-space model,

$$y_t = \mu_t + \varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (7)$$

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_\eta^2), \quad (8)$$

with the disturbance variances estimated on an initial 60-month window and then held fixed while the filter is run recursively. Taken together, HP, LOESS, and Kalman span standard linear trend extraction, local nonparametric smoothing, and linear Gaussian state-space filtering.

Table 3: Signal-Extraction Methods and Their Roles

Method	Family	Role
Raw	None	Baseline
HP	Penalized trend	Benchmark
LOESS	Local regression	Benchmark
Kalman	State-space	Benchmark
Savitzky-Golay	Local polynomial	Proposed
Perona-Malik	Nonlinear diffusion	Proposed

Together, these benchmark methods span the principal families of real-time smoothing approaches commonly used in econometrics, statistics, and engineering.

4.6 Parameter selection

The baseline hyperparameters are chosen from coarse grids by a parsimonious ex ante rule rather than by maximizing the final headline return. For Savitzky-Golay, the grid contains the three pairs (7, 2), (9, 2), and (11, 2). For Perona-Malik, the grid varies the iteration count and edge parameter around practical values used in the engineering literature. Within each family, the analysis first identifies the specifications whose Sharpe ratios are within 10% of the family maximum under the baseline economic design. Among that subset, it selects the lowest-turnover specification. This rule chooses $(h, p) = (9, 2)$ for Savitzky-Golay and $(\text{iterations}, \kappa) = (25, 2.0)$ for Perona-Malik. The full coarse-grid results are reported in the appendix.

4.7 Score construction and regime classification

After smoothing, each indicator is standardized with a trailing 48-month z -score,

$$z_{j,t} = \frac{\tilde{x}_{j,t} - \mu_{j,t}^{(48)}}{\sigma_{j,t}^{(48)}}, \quad (9)$$

and the composite macro score is the equal-weight average across the available indicators,

$$s_t = \frac{1}{N_t} \sum_{j=1}^{N_t} z_{j,t}. \quad (10)$$

The continuous score is mapped into three regimes using expanding quantile thresholds. At each month, observations below the 33rd percentile of the score history observed so far are labeled *stress*, observations above the 67th percentile are labeled *expansion*, and the remainder are labeled *slowdown*. Because both the normalization window and the regime thresholds use only information available up to date t , the procedure remains fully real-time throughout.

4.8 Allocation rule

At each formation month, regime-conditional mean sector returns are estimated from the trailing 60 months of ETF returns. The baseline portfolio then allocates to the single sector with the highest conditional mean return for the current regime and holds that position for one month. The top-one rule is used deliberately as a measurement device: if filtering improves the discriminatory content of the regime score, that improvement should be most visible when the allocation is concentrated rather than diversified away. Top-two and top-three allocations are reported as robustness checks.

4.9 Inference, multiple testing, and economic diagnostics

Three features of the empirical design are central for interpretation. First, each filter is one-sided, the regime thresholds expand recursively, and portfolio weights are formed exclusively from information available at the formation date. Second, all main cross-method comparisons are reported on the common sample shared by the candidate filters. Third, inference is based on paired monthly return differences together with dependence-robust bootstrap procedures.

The main significance table reports paired t -tests, 12-month moving-block bootstrap intervals, and stationary-bootstrap intervals. To address the concern that the reported baseline may sit inside a broader exploratory search, the analysis additionally defines a prespecified family of 249 candidate rules that varies the panel, the z -window, the training window, the portfolio breadth, and the local filter parameters. Family-wise evidence is then evaluated with White’s Reality Check (White, 2000), Hansen’s Superior Predictive Ability test (Hansen, 2005), the Deflated Sharpe Ratio (Bailey and Lopez de Prado, 2014), and the Probability of Backtest Overfitting (Bailey et al., 2017). These diagnostics are reported separately relative to the raw timing rule and the equal-weight sector

benchmark.

Transaction costs are modeled as 15 basis points per unit of one-way turnover. Performance is summarized with annualized return, annualized volatility, Sharpe ratio, maximum drawdown, turnover, regime-switch diagnostics, paired monthly return spreads, and multiple-testing adjustments. The methodological objective is therefore not merely to document higher returns, but to determine whether those returns survive real-time implementation, benchmark discipline, and explicit data-snooping controls.

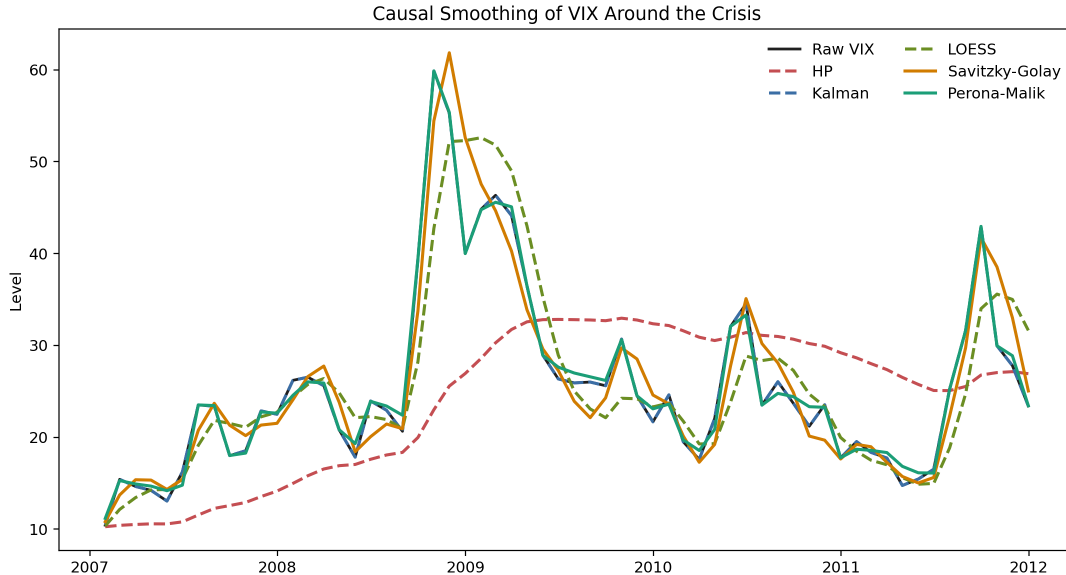


Figure 1: Real-time smoothing of the VIX around the global financial crisis. The raw VIX is shown in black, the proposed edge-preserving filters in solid color, and the benchmark smoothers in dashed lines. Perona-Malik retains the crisis spike more tightly than the other filters while still attenuating short-run variation.

5 Results

5.1 Main performance comparison

Table 4 reports the central performance comparison. The raw regime strategy performs poorly, with a 5.60% annualized return, 18.44% annualized volatility, a 0.304 Sharpe ratio, and a maximum drawdown of -53.1% . Every smoother improves on that baseline, but the magnitude of the improvement varies widely across methods.

Two patterns stand out. First, the proposed edge-preserving filters dominate the benchmark smoothers. Among the real-time filters considered, Perona-Malik delivers the strongest overall empirical performance: its annualized return reaches 15.85%, its Sharpe ratio reaches 0.820, and its return spread relative to the raw strategy exceeds 10 percentage points per year. Savitzky-Golay is the next strongest method at 13.78% and 0.707. Among the benchmarks, LOESS improves

materially on the raw signal but remains weaker than the proposed filters, while Kalman and recursive HP are economically helpful yet distinctly less competitive. The evidence is therefore not merely that “smoothing helps,” but that the form of smoothing matters.

Second, the equal-weight sector benchmark remains a demanding hurdle. It earns 10.57% with a 0.739 Sharpe ratio. Perona-Malik exceeds that benchmark economically on both dimensions, but the gap is not large enough to support an unconditional dominance claim. Table 4 therefore points to a clearer conclusion about improved regime extraction than about displacement of passive sector exposure.

Table 4: Main Performance Comparison

Method	Annual return	Annual volatility	Sharpe	Max drawdown	Return diff. vs. raw	Sharpe diff. vs. raw	Return diff. vs. benchmark	Sharpe diff. vs. benchmark
Equal weight	0.106	0.143	0.739	-0.491			0.000	0.000
Raw	0.056	0.184	0.304	-0.531	0.000	0.000	-0.050	-0.436
HP	0.097	0.194	0.502	-0.561	0.041	0.198	-0.008	-0.238
Kalman	0.088	0.182	0.485	-0.531	0.032	0.181	-0.017	-0.254
LOESS	0.125	0.195	0.638	-0.531	0.069	0.335	0.019	-0.101
Savitzky-Golay	0.138	0.195	0.707	-0.531	0.082	0.403	0.032	-0.033
Perona-Malik	0.158	0.193	0.820	-0.531	0.102	0.517	0.053	0.081

5.2 Paired statistical evidence and model confidence

Table 5 reports paired monthly return-difference tests. Perona-Malik minus raw delivers a 9.50 percentage-point annualized mean spread, a paired t -statistic of 3.38, and positive moving-block and stationary-bootstrap confidence intervals. Savitzky-Golay minus raw is also strongly positive, though less so than Perona-Malik. Relative to the modern comparators, Perona-Malik separates clearly from the Kalman filter and remains economically stronger than LOESS, even if the paired evidence against LOESS is less sharp.

The benchmark comparison requires more discipline in interpretation. Perona-Malik minus equal weight produces a 5.56 percentage-point annualized spread. The conventional two-sided paired t -test yields a p -value of 0.070, so the passive benchmark is not rejected at conventional two-sided levels. At the same time, both the moving-block and stationary one-sided bootstraps remain supportive. That combination of results is best read as economically favorable, but statistically less decisive than the comparison with the raw timing rule.

Table 5: Paired Monthly Return Differences

Comparison	Months	start_date	end_date	Mean monthly diff.	Annualized mean diff.	t-stat	Two-sided p-value	Bootstrap CI low	Bootstrap CI high	Bootstrap p-value	Stationary CI low	Stationary CI high	Stationary p-value
Savitzky-Golay minus raw	270	2004-03-29	2026-07-31	0.006	0.077	2.759	0.006	0.019	0.129	0.001	0.024	0.136	0.002
Perona-Malik minus raw	270	2004-03-29	2026-07-31	0.008	0.095	3.376	0.001	0.034	0.164	0.000	0.043	0.149	0.000
HP minus raw	270	2004-03-29	2026-07-31	0.003	0.040	1.275	0.203	-0.017	0.106	0.116	-0.020	0.105	0.092
LOESS minus raw	270	2004-03-29	2026-07-31	-0.003	0.066	1.914	0.012	0.010	0.152	0.011	0.003	0.140	0.018
Kalman minus raw	270	2004-03-29	2026-07-31	0.003	0.030	1.488	0.138	-0.006	0.063	0.053	-0.000	0.064	0.015
Perona-Malik minus Savitzky-Golay	270	2004-03-29	2026-07-31	0.001	0.018	0.709	0.425	-0.008	0.075	0.052	-0.036	0.063	0.234
Perona-Malik minus LOESS	270	2004-03-29	2026-07-31	0.002	0.029	1.279	0.202	-0.010	0.071	0.072	-0.011	0.068	0.076
Perona-Malik minus Kalman	270	2004-03-29	2026-07-31	0.005	0.065	2.696	0.007	0.024	0.137	0.001	0.008	0.128	0.014
Perona-Malik minus benchmark	270	2004-03-29	2026-07-31	0.005	0.056	1.816	0.070	0.014	0.116	0.004	0.010	0.102	0.011
Savitzky-Golay minus benchmark	270	2004-03-29	2026-07-31	0.003	0.038	1.180	0.239	-0.025	0.083	0.150	-0.016	0.098	0.097
LOESS minus benchmark	270	2004-03-29	2026-07-31	0.002	0.026	0.857	0.392	-0.022	0.091	0.121	-0.023	0.079	0.165
Kalman minus benchmark	270	2004-03-29	2026-07-31	-0.001	-0.009	-0.312	0.755	-0.062	0.042	0.706	-0.059	0.042	0.655

Table 6 addresses the data-snooping concern directly. The candidate family contains 249 strategies spanning panel choice, regime-standardization windows, training windows, portfolio breadth,

and local filter-parameter grids. Relative to the raw baseline, the White Reality Check and Superior Predictive Ability tests produce p -values of 0.062 and 0.076, respectively. These p -values are marginal and should not be overstated. They provide suggestive evidence that the observed improvement is unlikely to be solely attributable to unrestricted model search. Relative to the equal-weight benchmark, by contrast, the family-level multiple-testing evidence is weak. The Deflated Sharpe Ratio and the Probability of Backtest Overfitting tell the same story: the raw-comparison claim survives a meaningful selection penalty more convincingly than the benchmark-comparison claim.

Table 6: Multiple-Testing and Model-Confidence Diagnostics

Comparison	Candidate count	Months	Selected specification	Best family specification	Best family annualized diff.	White RC p -value	SPA p -value	Deflated Sharpe ratio	PBO
Candidate family minus raw baseline	249	210	PM $k=2.0$, $l=25$, full panel, $z=48$, $\text{train}=60$, $\text{top}=1$	PM $k=1.5$, $l=25$, without XLF/XLU, $z=36$, $\text{train}=48$, $\text{top}=1$	0.108	0.062	0.076	0.508	0.457
Candidate family minus equal-weight benchmark	249	210	PM $k=2.0$, $l=25$, full panel, $z=48$, $\text{train}=60$, $\text{top}=1$	PM $k=1.5$, $l=25$, without XLF/XLU, $z=36$, $\text{train}=48$, $\text{top}=1$	0.036	0.470	0.708	0.302	0.414

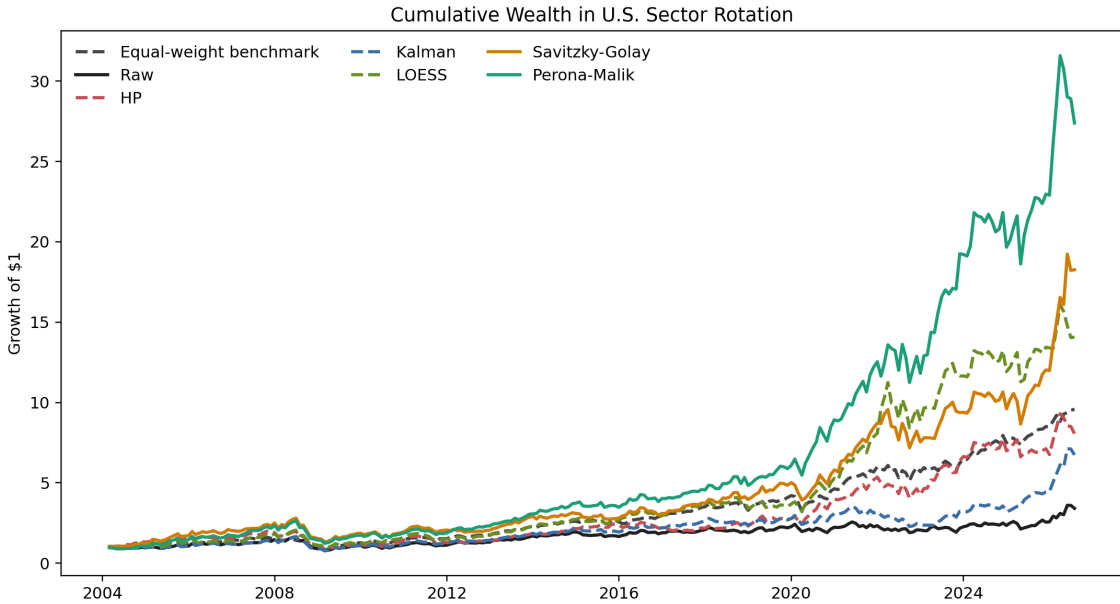


Figure 2: Cumulative wealth from February 2004 through July 2026. The proposed edge-preserving filters are shown in solid color, whereas the benchmark smoothers and the equal-weight benchmark are shown with dashed lines. The raw rule lags well behind the filtered variants, while Perona-Malik produces the strongest long-run compounding path among the real-time smoothers.

5.3 Regime mechanics and subperiod behavior

The mechanism evidence supports the economic interpretation of the return results. Table 7 shows that the raw signal switches regimes about 3.03 times per year. Perona-Malik reduces that rate to 1.82, while Savitzky-Golay reduces it to 2.56. Economically, fewer spurious regime transitions should improve performance because they reduce unnecessary portfolio turnover and stabilize the conditional return estimates used to map regimes into sector weights. At the same time, Perona-Malik cuts the share of high-frequency spectral power from 0.127 under the raw score to 0.058 without collapsing into the extreme inertia induced by the recursive HP benchmark. LOESS occu-

pies an intermediate position, whereas the Kalman benchmark remains much closer to the raw signal in both switching frequency and spectral profile. These diagnostics are consistent with the view that Perona-Malik improves the signal by removing regime noise rather than by indiscriminately flattening the series.

Table 7: Regime Stability and Frequency Diagnostics

Method	Regime switches per year	Avg. regime duration (months)	Median regime duration (months)	High-frequency power share	12-60m power share
Raw	3.032	3.904	2.000	0.127	0.365
HP	0.598	18.733	14.000	0.005	0.174
Kalman	2.947	4.014	2.000	0.110	0.371
LOESS	1.879	6.244	4.000	0.073	0.439
Savitzky-Golay	2.556	4.617	3.000	0.117	0.393
Perona-Malik	1.823	6.432	2.000	0.058	0.374

Figure 3 complements that interpretation in the time domain. Perona-Malik sustains a superior 60-month rolling Sharpe ratio through most of the post-2009 sample, whereas the raw strategy rarely approaches the filtered alternatives. The rolling evidence is therefore consistent with the full-sample and subperiod results.

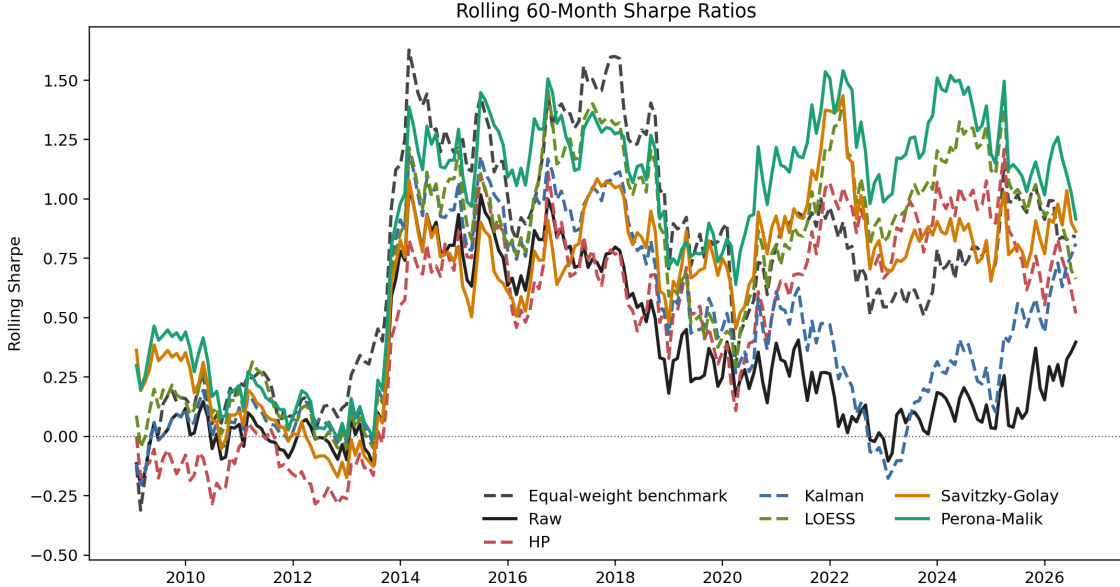


Figure 3: Rolling 60-month Sharpe ratios. Solid lines denote the proposed edge-preserving filters and dashed lines denote benchmark methods. Perona-Malik remains persistently stronger than the raw rule and usually stronger than the benchmark smoothers over the post-2009 sample.

Figure 4 now separates the proposed filters from the benchmark smoothers to make their distinct signatures easier to see. The left column reports Savitzky-Golay and Perona-Malik, while the right column reports HP, LOESS, and Kalman. Within each column, the top panel shows score levels and the bottom panel shows deviations from the raw score. Savitzky-Golay and especially Perona-Malik remain closer to the raw turning points than the HP benchmark, but they are less erratic than the

unsmoothed score. LOESS and Kalman occupy intermediate positions. This pattern is consistent with a regime filter that attenuates noise without imposing excessive delay.

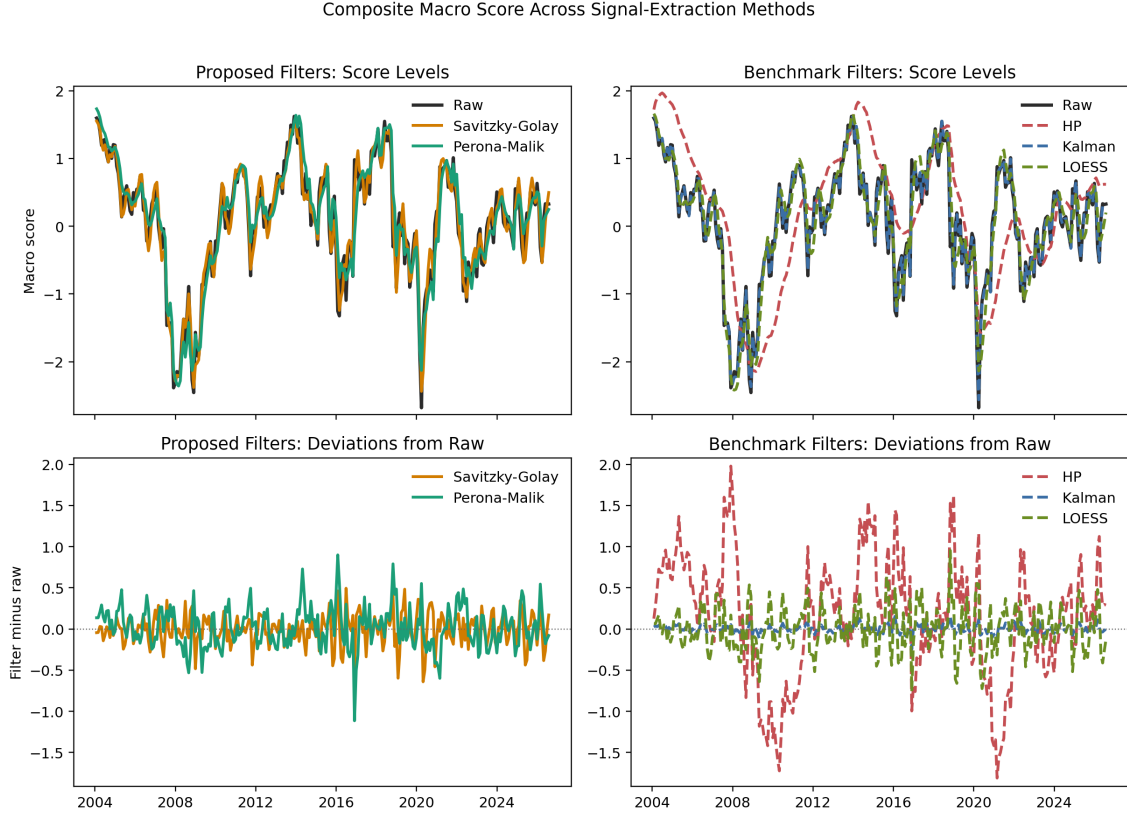


Figure 4: Composite macro scores across methods. The left column reports the proposed edge-preserving filters and the right column reports benchmark smoothers. In each column, the top panel shows score levels and the bottom panel shows deviations from the raw score. Perona-Malik suppresses short-run score noise more than the raw and Kalman filters while remaining less inertial than recursive HP.

Subperiod evidence in Table 8 shows that the gains are not confined to one isolated episode, although they are not uniform across all market environments. The relative advantage of the filtered rules is strongest after 2009 and especially after 2020, whereas the 2004–2008 interval is noisier. That pattern is economically plausible: the value of a better regime filter should be largest when cyclical-versus-defensive leadership changes sharply but does not become entirely chaotic.

Table 8: Subperiod Performance

Method	Subperiod	Annual return	Annual volatility	Sharpe	Max drawdown
Equal weight	2004–2008	-0.000	0.129	-0.001	-0.385
Equal weight	2009–2014	0.173	0.149	1.166	-0.159
Equal weight	2015–2019	0.100	0.115	0.873	-0.124
Equal weight	2020–2026	0.133	0.164	0.812	-0.226
Raw	2004–2008	-0.021	0.214	-0.100	-0.456
Raw	2009–2014	0.137	0.165	0.832	-0.163
Raw	2015–2019	0.031	0.127	0.248	-0.146
Raw	2020–2026	0.064	0.213	0.298	-0.260
HP	2004–2008	0.016	0.200	0.080	-0.457
HP	2009–2014	0.120	0.191	0.630	-0.267
HP	2015–2019	0.054	0.151	0.356	-0.202
HP	2020–2026	0.176	0.220	0.798	-0.245
Kalman	2004–2008	-0.021	0.214	-0.100	-0.456
Kalman	2009–2014	0.153	0.162	0.946	-0.162
Kalman	2015–2019	0.056	0.124	0.453	-0.146
Kalman	2020–2026	0.144	0.210	0.686	-0.356
LOESS	2004–2008	0.023	0.222	0.105	-0.456
LOESS	2009–2014	0.161	0.168	0.958	-0.247
LOESS	2015–2019	0.061	0.146	0.417	-0.164
LOESS	2020–2026	0.226	0.227	0.995	-0.251
Savitzky-Golay	2004–2008	0.090	0.235	0.384	-0.456
Savitzky-Golay	2009–2014	0.125	0.161	0.777	-0.176
Savitzky-Golay	2015–2019	0.101	0.135	0.750	-0.133
Savitzky-Golay	2020–2026	0.217	0.228	0.951	-0.249
Perona-Malik	2004–2008	0.076	0.240	0.317	-0.456
Perona-Malik	2009–2014	0.173	0.157	1.101	-0.163
Perona-Malik	2015–2019	0.102	0.132	0.772	-0.101
Perona-Malik	2020–2026	0.257	0.221	1.161	-0.174

5.4 Robustness, parameter sensitivity, and trading activity

The robustness results sharpen, rather than overturn, the baseline interpretation. Table 9 shows that the filtered strategies remain ahead of the raw rule across multiple training-window choices, with Perona-Malik particularly resilient. The parameter-selection appendix is also informative: the baseline Savitzky-Golay and Perona-Malik calibrations are not isolated maxima chosen for return alone. Instead, they are selected by a coarse-grid parsimony rule that retains low-turnover specifications within 10% of the family-best Sharpe ratio.

Bootstrap sensitivity checks in the appendix further reduce the concern that the inference depends on one arbitrary block length. The conclusion that Perona-Malik dominates the raw rule is stable across 6-, 12-, and 18-month moving-block bootstraps as well as a stationary bootstrap. The comparison with the equal-weight benchmark is weaker but still remains economically positive across those alternatives. More generally, however, the sample contains 270 monthly observations, so the paper should be read as providing disciplined evidence rather than high-power proof on every comparison of interest.

Trading evidence is consistent with the main return results rather than in tension with them. Table 10 shows that the raw top-one strategy turns over 8.18 times per year on average. Savitzky-Golay lowers that figure to 6.22, and Perona-Malik lowers it further to 4.36. More than 80% of months involve zero turnover under the Perona-Malik baseline, which is consistent with a filter that stabilizes the state estimate and reduces spurious regime reclassification. The appendix confirms that those turnover reductions are economically meaningful relative to the raw rule and to Savitzky-Golay.

Taken together, the robustness exercises support a disciplined conclusion. The paper does not show that a filtered sector-rotation rule unambiguously dominates passive sector exposure under every inferential lens. It does show, however, that real-time edge-preserving filtering materially improves the extraction of an implementable regime signal relative to the raw panel and relative to more standard benchmark smoothers, and that this conclusion survives a nontrivial battery of multiple-testing and sensitivity checks.

Table 9: Training-Window Robustness

family	Value	Method	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	Return diff. vs. raw	Sharpe diff. vs. raw	Return diff. vs. benchmark	Sharpe diff. vs. benchmark
training_window	48.000	Raw	282	2003-02-28	2026-07-31	0.071	0.186	0.380	-0.531	0.000	0.000	-0.044	-0.429
training_window	48.000	Savitzky-Golay	276	2003-08-31	2026-07-31	0.126	0.195	0.644	-0.531	0.057	0.279	0.016	-0.132
training_window	48.000	Perona-Malik	282	2003-02-28	2026-07-31	0.131	0.196	0.668	-0.531	0.060	0.288	0.016	-0.141
training_window	48.000	HP	280	2003-04-30	2026-07-31	0.099	0.191	0.520	-0.561	0.027	0.132	-0.017	-0.295
training_window	60.000	Raw	270	2004-02-29	2026-07-31	0.056	0.184	0.304	-0.531	0.000	0.000	-0.050	-0.436
training_window	60.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.138	0.195	0.707	-0.531	0.082	0.403	0.032	-0.033
training_window	60.000	Perona-Malik	270	2004-02-29	2026-07-31	0.138	0.193	0.820	-0.531	0.102	0.517	0.053	0.081
training_window	60.000	HP	270	2004-02-29	2026-07-31	0.097	0.194	0.502	-0.561	0.041	0.198	-0.008	-0.238
training_window	84.000	Raw	246	2006-02-28	2026-07-31	0.083	0.183	0.452	-0.531	0.000	0.000	-0.021	-0.252
training_window	84.000	Savitzky-Golay	246	2006-02-28	2026-07-31	0.124	0.200	0.621	-0.531	0.041	0.169	0.020	-0.083
training_window	84.000	Perona-Malik	246	2006-02-28	2026-07-31	0.136	0.198	0.780	-0.531	0.073	0.337	0.052	0.085
training_window	84.000	HP	246	2006-02-28	2026-07-31	0.103	0.202	0.500	-0.561	0.020	0.057	-0.001	-0.195
training_window	120.000	Raw	210	2009-02-28	2026-07-31	0.085	0.176	0.482	-0.284	0.000	0.000	-0.028	-0.510
training_window	120.000	Savitzky-Golay	210	2009-02-28	2026-07-31	0.154	0.187	0.824	-0.393	0.069	0.342	0.011	-0.168
training_window	120.000	Perona-Malik	210	2009-02-28	2026-07-31	0.160	0.186	0.861	-0.307	0.075	0.380	0.017	-0.130
training_window	120.000	HP	210	2009-02-28	2026-07-31	0.140	0.198	0.707	-0.312	0.055	0.225	-0.003	-0.285

Table 10: Turnover and Trading Inactivity

Method	Average monthly turnover	Annualized turnover	Share of zero-turnover months
Raw	0.681	8.178	0.659
HP	0.430	5.156	0.785
Kalman	0.630	7.556	0.685
LOESS	0.444	5.333	0.778
Savitzky-Golay	0.519	6.222	0.741
Perona-Malik	0.363	4.356	0.819

6 Conclusion

This paper studies whether edge-preserving macro filtering improves a real-time U.S. sector-rotation rule constructed from public data and liquid sector ETFs. Relative to the unsmoothed regime signal, both Savitzky-Golay and Perona-Malik produce economically large improvements, and among the real-time filters considered Perona-Malik delivers the strongest overall empirical performance. It also outperforms the additional LOESS and Kalman benchmarks, indicating that the result is not simply a generic consequence of smoothing.

The benchmark comparison is more qualified. Perona-Malik exceeds the equal-weight sector benchmark economically, and the one-sided dependence-robust bootstrap evidence is favorable, but the conventional two-sided paired test is less decisive and the family-wise multiple-testing corrections are not strong enough to support an unconditional dominance claim over passive sector exposure. That distinction is central. The paper supports a strong conclusion about *improving regime extraction relative to the raw signal* and a more cautious conclusion about *replacing passive sector exposure*.

The multiple-testing diagnostics also sharpen the interpretation. Once the candidate family is defined explicitly, the White Reality Check and Superior Predictive Ability statistics remain only marginal for the raw comparison, while the evidence against the passive benchmark is materially weaker still. The Deflated Sharpe Ratio and the Probability of Backtest Overfitting point in the same direction. The argument therefore does not rely on the premise that the reported baseline is the unique best rule discovered after an unrestricted search, but it should also not be read as a fully closed case against data-mining concerns.

The mechanism evidence is equally important. Perona-Malik reduces regime-switch frequency, lowers short-horizon spectral power, and still avoids the excessive inertia of the recursive HP benchmark. In that sense, the filter appears to improve the state estimate by stabilizing economically irrelevant fluctuations while preserving the medium-horizon structure needed for allocation decisions.

For quantitative finance, the main implication is methodological. In regime-aware asset allocation, the filtering layer is part of the model, not merely a cosmetic preprocessing step. The U.S. sector-rotation evidence reported here suggests that real-time edge-preserving filters deserve explicit

consideration whenever the latent state is noisy, discontinuous, and economically consequential for portfolio choice. At the same time, the scope of the present evidence remains limited to one market, one ETF universe, and 270 monthly observations. The top-one portfolio rule is intentionally simple because it isolates the informational content of the regime signal, but that choice leaves open the separate question of how such signals would perform inside richer allocation schemes. Cross-market replication and alternative portfolio constructions therefore remain natural directions for subsequent work.

A Appendix

A.1 Design-search and model-selection details

Table 11 reports the strongest candidate specifications from the prespecified model family used in the multiple-testing corrections. The family is intentionally broad enough to include panel variation, alternative regime-standardization windows, alternative training windows, alternative portfolio breadth choices, and local parameter grids around the Savitzky-Golay and Perona-Malik baselines.

Table 11: Top Specifications from the Prespecified Candidate Family

panel	Method	Parameters	z_window	training_window	top_n	Annual return	Sharpe	Return diff. vs. raw
Without VIX	Savitzky-Golay	w=9, p=2	36	120	1	0.180	0.940	0.109
Full panel	Perona-Malik	iter=25, k=2.0	48	60	1	0.158	0.820	0.102
Full panel	Perona-Malik	iter=25, k=1.0	48	60	1	0.150	0.788	0.094
Full panel	Perona-Malik	iter=25, k=1.5	48	60	1	0.144	0.751	0.088
Full panel	Perona-Malik	iter=35, k=2.0	48	60	1	0.142	0.745	0.086
Full panel	Perona-Malik	iter=25, k=1.5	60	120	1	0.185	0.961	0.097
Full panel	Savitzky-Golay	w=9, p=2	48	60	1	0.138	0.707	0.082
Full panel	Perona-Malik	iter=15, k=1.0	48	60	1	0.126	0.675	0.070
Full panel	Perona-Malik	iter=25, k=1.5	60	60	1	0.152	0.755	0.080
Full panel	HP	lambda=129600	60	120	1	0.169	0.881	0.080

Table 12 reports the coarse-grid parameter-selection exercise. The baseline values retained in the main text are those selected by the parsimonious rule described in Section 4: within each filter family, keep the specifications whose Sharpe ratios lie within 10% of the family maximum and then choose the lowest-turnover member of that set.

Table 12: Coarse-Grid Parameter Selection

Method	Parameters	Annual return	Sharpe	Annualized turnover	Within 10% of best Sharpe	Selected by parsimony rule
Savitzky-Golay	w=7, p=2	0.056	0.309	7.467	No	No
Savitzky-Golay	w=9, p=2	0.138	0.707	6.222	Yes	Yes
Savitzky-Golay	w=11, p=2	0.120	0.613	6.578	No	No
Perona-Malik	iter=15, k=1.0	0.126	0.675	5.333	No	No
Perona-Malik	iter=25, k=1.0	0.150	0.788	4.711	Yes	No
Perona-Malik	iter=25, k=1.5	0.144	0.751	4.622	Yes	No
Perona-Malik	iter=25, k=2.0	0.158	0.820	4.356	Yes	Yes
Perona-Malik	iter=35, k=2.0	0.142	0.745	4.800	Yes	No

A.2 Additional robustness

Table 13: Regime Standardization Window Robustness

family	Value	Method	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	Return diff. vs. raw	Sharpe diff. vs. raw	Return diff. vs. benchmark	Sharpe diff. vs. benchmark
z_window	36.000	Raw	270	2004-02-29	2026-07-31	0.115	0.190	0.603	-0.531	0.000	0.000	0.009	-0.136
z_window	36.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.112	0.187	0.598	-0.531	-0.003	-0.005	0.006	-0.141
z_window	36.000	Perona-Malik	270	2004-02-29	2026-07-31	0.150	0.197	0.759	-0.531	0.035	0.156	0.044	0.020
z_window	36.000	HP	270	2004-02-29	2026-07-31	0.108	0.198	0.543	-0.531	-0.007	-0.060	0.002	-0.196
z_window	48.000	Raw	270	2004-02-29	2026-07-31	0.056	0.184	0.304	-0.531	0.000	0.000	-0.050	-0.436
z_window	48.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.138	0.195	0.707	-0.531	0.082	0.403	0.032	-0.033
z_window	48.000	Perona-Malik	270	2004-02-29	2026-07-31	0.158	0.193	0.820	-0.531	0.102	0.517	0.053	0.081
z_window	48.000	HP	270	2004-02-29	2026-07-31	0.097	0.194	0.502	-0.561	0.041	0.198	-0.008	-0.238
z_window	60.000	Raw	270	2004-02-29	2026-07-31	0.072	0.181	0.396	-0.531	0.000	0.000	-0.034	-0.344
z_window	60.000	Savitzky-Golay	264	2004-08-31	2026-07-31	0.101	0.192	0.524	-0.531	0.028	0.127	-0.007	-0.224
z_window	60.000	Perona-Malik	270	2004-02-29	2026-07-31	0.151	0.199	0.757	-0.531	0.079	0.362	0.045	0.018
z_window	60.000	HP	268	2004-04-30	2026-07-31	0.101	0.193	0.523	-0.531	0.031	0.136	-0.005	-0.217

Table 14: Transaction-Cost Robustness

family	Value	Method	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	Return diff. vs. raw	Sharpe diff. vs. raw	Return diff. vs. benchmark	Sharpe diff. vs. benchmark
transaction_cost_bps	0.000	Raw	270	2004-02-29	2026-07-31	0.069	0.185	0.373	-0.531	0.000	0.000	-0.037	-0.366
transaction_cost_bps	0.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.148	0.195	0.762	-0.531	0.079	0.388	0.043	0.022
transaction_cost_bps	0.000	Perona-Malik	270	2004-02-29	2026-07-31	0.166	0.193	0.859	-0.531	0.097	0.486	0.060	0.120
transaction_cost_bps	0.000	HP	270	2004-02-29	2026-07-31	0.106	0.194	0.545	-0.557	0.037	0.172	0.000	-0.194
transaction_cost_bps	15.000	Raw	270	2004-02-29	2026-07-31	0.056	0.184	0.304	-0.531	0.000	0.000	-0.050	-0.436
transaction_cost_bps	15.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.138	0.195	0.707	-0.531	0.082	0.403	0.032	-0.033
transaction_cost_bps	15.000	Perona-Malik	270	2004-02-29	2026-07-31	0.158	0.193	0.820	-0.531	0.102	0.517	0.053	0.081
transaction_cost_bps	15.000	HP	270	2004-02-29	2026-07-31	0.097	0.194	0.502	-0.561	0.041	0.198	-0.008	-0.238
transaction_cost_bps	30.000	Raw	270	2004-02-29	2026-07-31	0.043	0.184	0.235	-0.531	0.000	0.000	-0.062	-0.505
transaction_cost_bps	30.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.127	0.195	0.652	-0.531	0.084	0.417	0.022	-0.087
transaction_cost_bps	30.000	Perona-Malik	270	2004-02-29	2026-07-31	0.151	0.193	0.782	-0.531	0.108	0.547	0.045	0.042
transaction_cost_bps	30.000	HP	270	2004-02-29	2026-07-31	0.089	0.194	0.458	-0.568	0.046	0.223	-0.017	-0.281
transaction_cost_bps	50.000	Raw	270	2004-02-29	2026-07-31	0.026	0.184	0.143	-0.531	0.000	0.000	-0.079	-0.597
transaction_cost_bps	50.000	Savitzky-Golay	270	2004-02-29	2026-07-31	0.113	0.196	0.579	-0.531	0.087	0.436	0.008	-0.161
transaction_cost_bps	50.000	Perona-Malik	270	2004-02-29	2026-07-31	0.141	0.193	0.730	-0.531	0.115	0.587	0.035	-0.010
transaction_cost_bps	50.000	HP	270	2004-02-29	2026-07-31	0.078	0.194	0.400	-0.577	0.051	0.257	-0.028	-0.340

Table 15: Panel Sensitivity

family	Value	Method	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	Return diff. vs. raw	Sharpe diff. vs. raw	Return diff. vs. benchmark	Sharpe diff. vs. benchmark
panel_variant	full_panel	Raw	270	2004-02-29	2026-07-31	0.056	0.184	0.304	-0.531	0.000	0.000	-0.050	-0.436
panel_variant	full_panel	Savitzky-Golay	270	2004-02-29	2026-07-31	0.138	0.195	0.707	-0.531	0.082	0.403	0.032	-0.033
panel_variant	full_panel	Perona-Malik	270	2004-02-29	2026-07-31	0.158	0.193	0.820	-0.531	0.102	0.517	0.053	0.081
panel_variant	full_panel	HP	270	2004-02-29	2026-07-31	0.097	0.194	0.502	-0.561	0.041	0.198	-0.008	-0.238
panel_variant	drop_vix	Raw	270	2004-02-29	2026-07-31	0.117	0.197	0.595	-0.531	0.000	0.000	0.011	-0.145
panel_variant	drop_vix	Savitzky-Golay	270	2004-02-29	2026-07-31	0.107	0.197	0.542	-0.531	-0.010	-0.053	0.001	-0.198
panel_variant	drop_vix	Perona-Malik	270	2004-02-29	2026-07-31	0.105	0.194	0.543	-0.531	-0.012	-0.052	-0.000	-0.196
panel_variant	drop_vix	HP	270	2004-02-29	2026-07-31	0.121	0.202	0.601	-0.531	0.004	0.006	0.016	-0.139
panel_variant	drop_russell_large_ratio	Raw	270	2004-02-29	2026-07-31	0.096	0.183	0.524	-0.531	0.000	0.000	-0.010	-0.215
panel_variant	drop_russell_large_ratio	Savitzky-Golay	270	2004-02-29	2026-07-31	0.100	0.195	0.515	-0.531	0.004	-0.010	-0.005	-0.225
panel_variant	drop_russell_large_ratio	Perona-Malik	270	2004-02-29	2026-07-31	0.111	0.197	0.562	-0.531	0.015	0.038	0.005	-0.177
panel_variant	drop_russell_large_ratio	HP	270	2004-02-29	2026-07-31	0.079	0.198	0.397	-0.560	-0.017	-0.127	-0.027	-0.342
panel_variant	drop_xly_xly_ratio	Raw	270	2004-02-29	2026-07-31	0.074	0.183	0.403	-0.534	0.000	0.000	-0.032	-0.337
panel_variant	drop_xly_xly_ratio	Savitzky-Golay	270	2004-02-29	2026-07-31	0.100	0.193	0.517	-0.531	0.026	0.114	-0.006	-0.223
panel_variant	drop_xly_xly_ratio	Perona-Malik	270	2004-02-29	2026-07-31	0.086	0.186	0.460	-0.531	0.012	0.057	-0.020	-0.280
panel_variant	drop_xly_xly_ratio	HP	270	2004-02-29	2026-07-31	0.094	0.195	0.482	-0.531	0.020	0.079	-0.011	-0.257
panel_variant	drop_xlf_xlf_ratio	Raw	270	2004-02-29	2026-07-31	0.106	0.181	0.386	-0.497	0.000	0.000	0.001	-0.154
panel_variant	drop_xlf_xlf_ratio	Savitzky-Golay	270	2004-02-29	2026-07-31	0.090	0.175	0.515	-0.430	-0.016	-0.071	-0.016	-0.215
panel_variant	drop_xlf_xlf_ratio	Perona-Malik	270	2004-02-29	2026-07-31	0.085	0.185	0.461	-0.527	-0.021	-0.125	-0.020	-0.279
panel_variant	drop_xlf_xlf_ratio	HP	270	2004-02-29	2026-07-31	0.085	0.192	0.444	-0.531	-0.021	-0.142	-0.020	-0.296

Table 16: Bootstrap Sensitivity

Comparison	Bootstrap method	Annualized mean diff.	Bootstrap CI low	Bootstrap CI high	Bootstrap p-value
Perona-Malik minus raw	Moving block (6)	0.095	0.054	0.159	0.000
Perona-Malik minus raw	Moving block (12)	0.095	0.054	0.164	0.000
Perona-Malik minus raw	Moving block (18)	0.095	0.052	0.158	0.000
Perona-Malik minus raw	Stationary (12)	0.095	0.043	0.149	0.000
Perona-Malik minus benchmark	Moving block (6)	0.056	0.013	0.116	0.005
Perona-Malik minus benchmark	Moving block (12)	0.056	0.014	0.116	0.004
Perona-Malik minus benchmark	Moving block (18)	0.056	0.010	0.110	0.007
Perona-Malik minus benchmark	Stationary (12)	0.056	0.010	0.102	0.011

Table 17: Turnover-Difference Tests

Comparison	Mean monthly diff.	Annualized mean diff.	Stationary CI low	Stationary CI high	Stationary p-value
Perona-Malik minus raw	-0.319	-3.822	-5.778	-1.956	1.000
Perona-Malik minus HP	-0.067	-0.800	-3.289	1.778	0.742
Perona-Malik minus Kalman	-0.267	-3.200	-5.067	-1.511	1.000
Perona-Malik minus LOESS	-0.081	-0.978	-2.311	0.267	0.937
Perona-Malik minus Savitzky-Golay	-0.156	-1.867	-3.289	-0.444	0.995

A.3 Parameter sensitivity

Table 18: Savitzky-Golay Parameter Sensitivity

Method	Value	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	annual_return_diff_vs_default	sharpe_diff_vs_default	Return diff. vs. raw	Sharpe diff. vs. raw
Savitzky-Golay	w=7, p=2	270	2004-02-29	2026-07-31	0.056	0.182	0.309	-0.526	-0.081	-0.398	0.000	0.005
Savitzky-Golay	w=9, p=2	270	2004-02-29	2026-07-31	0.138	0.195	0.707	-0.531	0.000	0.000	0.082	0.403
Savitzky-Golay	w=11, p=2	270	2004-02-29	2026-07-31	0.120	0.197	0.613	-0.531	-0.017	-0.094	0.064	0.309

Table 19: Perona-Malik Parameter Sensitivity

Method	Value	Months	start_date	end_date	Annual return	Annual volatility	Sharpe	Max drawdown	annual_return_diff_vs_default	sharpe_diff_vs_default	Return diff. vs. raw	Sharpe diff. vs. raw
Perona-Malik	iter=15, k=1.0	270	2004-02-29	2026-07-31	0.126	0.186	0.675	-0.531	-0.033	-0.145	0.070	0.371
Perona-Malik	iter=25, k=1.0	270	2004-02-29	2026-07-31	0.150	0.191	0.788	-0.531	-0.008	-0.033	0.094	0.484
Perona-Malik	iter=25, k=1.5	270	2004-02-29	2026-07-31	0.144	0.191	0.751	-0.531	-0.015	-0.069	0.088	0.447
Perona-Malik	iter=25, k=2.0	270	2004-02-29	2026-07-31	0.158	0.193	0.820	-0.531	0.000	0.000	0.102	0.517
Perona-Malik	iter=35, k=2.0	270	2004-02-29	2026-07-31	0.142	0.191	0.745	-0.531	-0.017	-0.076	0.086	0.441

References

- David H. Bailey and Marcos Lopez de Prado. The deflated sharpe ratio: Correcting for selection bias, backtest overfitting and non-normality. *Journal of Portfolio Management*, 40(5):94–107, 2014.
- David H. Bailey, Jonathan Borwein, Marcos Lopez de Prado, and Qiji Jim Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69, 2017.
- John Y. Campbell and Samuel B. Thompson. Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531, 2008.
- William S. Cleveland. Robust locally weighted regression and smoothing scatterplots. *Journal of the American Statistical Association*, 74(368):829–836, 1979.
- Victor DeMiguel, Lorenzo Garlappi, and Raman Uppal. Optimal versus naive diversification: How inefficient is the 1/n portfolio strategy? *Review of Financial Studies*, 22(5):1915–1953, 2009.
- Eugene F. Fama and Kenneth R. French. Business conditions and expected returns on stocks and bonds. *Journal of Financial Economics*, 25(1):23–49, 1989.
- Wayne E. Ferson and Campbell R. Harvey. The variation of economic risk premiums. *Journal of Political Economy*, 99(2):385–415, 1991.

- Peter R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.
- Robert J. Hodrick and Edward C. Prescott. Postwar u.s. business cycles: An empirical investigation. *Journal of Money, Credit and Banking*, 29(1):1–16, 1997.
- Rudolf E. Kalman. A new approach to linear filtering and prediction problems. *Journal of Basic Engineering*, 82(1):35–45, 1960.
- Tobias J. Moskowitz, Yao Hua Ooi, and Lasse Heje Pedersen. Time series momentum. *Journal of Financial Economics*, 104(2):228–250, 2012.
- Pietro Perona and Jitendra Malik. Scale-space and edge detection using anisotropic diffusion. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 12(7):629–639, 1990. doi: 10.1109/34.56205.
- Abraham Savitzky and Marcel J. E. Golay. Smoothing and differentiation of data by simplified least squares procedures. *Analytical Chemistry*, 36(8):1627–1639, 1964. doi: 10.1021/ac60214a047.
- Ivo Welch and Amit Goyal. A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508, 2008.
- Halbert White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000.